

Kgarira

Repository Documentation

Current technical reference for the React + Django ticketing application.

Snapshot	Value
Repository	kgarira
Document date	2026-09-09
Frontend	React 19, Vite 8, Tailwind CSS 4, React Router, Axios
Backend	Django 6.1, Django REST Framework, SQLite
Payments	eSewa sandbox initiation; partial verification screen

Purpose

This document records the repository as scanned and separates implemented behavior from unfinished or unsafe areas.

Status summary: Event browsing, guest booking creation, inventory locking, checkout state and eSewa initiation are present. eSewa verification does not yet confirm a booking. DEBUG and a Django secret remain in source, and there are no meaningful automated tests.

1. Architecture

React renders the catalogue and checkout. Django exposes REST endpoints, owns ticket inventory and creates signed eSewa payloads. In development, Browser -> Vite localhost:5173 -> proxy -> Django 127.0.0.1:8000. Relative API URLs use the Vite proxy when VITE_DJANGO_BASE_URL is absent.

Layer	Responsibility	Location
React client	Pages, routing, API calls and checkout UI	frontend/src/
Vite dev server	Dev server and routes proxy	frontend/vite.config.js
Django API	CRUD and eSewa endpoints	backend/TMS/tickets/
Persistence	SQLite and uploaded media	backend/TMS/
eSewa sandbox	Signed browser form target	rc-epay.esewa.com.np

Repository layout

Path	Contents
backend/TMS/TMS/	Settings, root URLs, ASGI/WSGI
backend/TMS/tickets/	Models, serializers, views, URLs, admin, migrations, payment utility
frontend/src/pages/	Route-level screens
frontend/src/components/	Site and checkout UI
frontend/src/context/	Checkout state provider
frontend/src/utlis/	Date format and eSewa redirect helper

2. Data model

Model	Key fields / behavior
Category	name; Event category is optional and is set null if deleted.
Event	title, unique slug, description, category, organizer, venue, dates, image, status, timestamps. Statuses are DRAFT, CANCELLED, PUBLISHED, COMPLETED. End date is constrained to be on/after start date; slugs auto-generate.
Ticket	event, REGULAR/VIP/VVIP type, quantity, sold_quantity, integer price.
Booking	ticket, quantity, PENDING/CONFIRMED/CANCELLED status, buyer fields, created timestamp, transaction_uuid; total_price is ticket price x quantity.

Inventory behavior

BookingSerializer locks the ticket in a transaction, checks quantity - sold_quantity, increments sold_quantity, then creates the booking. It prevents concurrent overselling but also consumes stock while a payment is still PENDING.

3. Backend API

Endpoint	Methods	Behavior
/events/	GET, POST	Event list/create
/events/{slug}/	GET, PUT, PATCH, DELETE	Slug-based event detail; includes nested tickets
/tickets/ and /tickets/{id}/	CRUD	Ticket endpoints
/category/ and /category/{id}/	CRUD	Category endpoints
/bookings/ and /bookings/{id}/	CRUD	Booking endpoints; current query exposes all bookings
/payments/esewa/initiate/	POST	Signed eSewa v2 payload for PENDING booking
/payments/esewa/verify/	POST	Decodes eSewa data only; no signature check or booking update

eSewa implementation

Initiation calculates formatted monetary fields, assigns transaction_uuid, signs total_amount, transaction_uuid and product_code with HMAC-SHA256/Base64, and returns {form_url, payload}. The frontend helper sends only eSewa fields as a native hidden HTML form POST, avoiding CORS restrictions. Settings load .env files and provide sandbox defaults.

4. Frontend routes

Route	Screen / behavior
/	HomePage with hero/list, categories and event grid
/book-venue	Static Book Venue page
/book-artist	Static Book Artist page
/contact	Static Contact page
/terms	Terms page
/buy-tickets/:slug	Ticket selection for fetched event
/buy-tickets/:slug/details	Buyer details and terms acceptance
/buy-tickets/:slug/payment	Payment selection and booking/eSewa initiation
/buy-tickets/:slug/payment/success	eSewa return screen; incomplete and mismatched with current success URL

5. Frontend components

Component	Current role
MainLayout / Navbar / Footer	Global shell. List Your Events has no action; some footer links have no matching routes.
EventList / EventCard	Home hero. EventList reads import.meta.env directly rather than api.js.
CategoryButtons	Fetches categories but active category is not shared with EventGrid, so it does not filter.
EventGrid	Fetches non-CANCELLED events, renders ticket prices and opens checkout; it also permits DRAFT/COMPLETED events.

Component	Current role
CheckoutLayout + CheckoutContext	Fetches event by slug and retains ticket, quantity, buyer fields and accepted terms across nested checkout.
TicketGrid	Selection and displayed total; quantity plus control has no availability cap.
Payment	Creates booking then starts eSewa. Selecting unimplemented Khalti still creates a booking.
PaymentSuccessful	Attempts verification but BASE_URL is not imported, error variable is wrong, and success state is never set.

6. Booking and payment flow

1. Event grid opens /buy-tickets/:slug.
2. Checkout layout GETs /events/:slug/ and passes the event to nested pages.
3. Buyer selects ticket/quantity and enters personal details.
4. Payment POSTs ticket, quantity and buyer fields to /bookings/.
5. Backend locks inventory, increments sold_quantity and creates a PENDING booking.
6. For eSewa, client POSTs booking_id to /payments/esewa/initiate/.
7. Django returns signed payload; browser submits it to sandbox as a form POST.
8. Sandbox returns data; current verify endpoint decodes it only and does not confirm booking.

Important mismatch

Default ESEWA_SUCCESS_URL is `http://localhost:5173/payment/success`, but React defines only `/buy-tickets/:slug/payment/success`. The redirect therefore has no matching client route.

7. Risks and next steps

Priority	Issue	Recommended work
High	Verification is not authoritative	Verify signature and amount/product, locate booking by transaction_uuid, and atomically confirm it exactly once.
High	Booking access is unrestricted	Replace all-bookings ModelViewSet exposure with authentication or scoped guest access tokens.
High	Pending bookings consume stock	Create expiring reservations or increment inventory only on verified confirmation.
High	Production configuration unsafe	Move secrets to env, disable DEBUG, set ALLOWED_HOSTS and gateway callback URLs.
Medium	Khalti creates abandoned booking	Validate method before booking creation or implement Khalti.
Medium	No tests	Add model, serializer, API, payment and e2e checkout tests.
Medium	Visibility/filtering gaps	Filter PUBLISHED events and connect category state to grid.
Low	Media URLs inconsistent	Use shared mediaUrl helper in checkout/cards.

Local development

Run Django from backend/TMS with `manage.py runserver` and Vite from frontend with `npm run dev`. The proxy expects Django at `127.0.0.1:8000`. Override gateway configuration in backend `.env` with `ESEWA_PRODUCT_CODE`, `ESEWA_SECRET_KEY`, `ESEWA_FORM_URL`, `ESEWA_SUCCESS_URL` and `ESEWA_FAILURE_URL`.

Conclusion

Kgarira has a solid event-catalogue and checkout foundation. The immediate production milestone is a secure, idempotent payment lifecycle that scopes booking access, verifies eSewa callbacks, reconciles stock once, and is covered by tests.